

Modules over PIDs

Observation:

R : integral domain with the following properties

(1) For any non-zero $r \in R$

r is irreducible

$(\Leftrightarrow) r$ is prime

(2) Every non-invertible & non-zero element can be written

as a product of irreducible

elements.

Then every non-invertible & non-zero element $r \in R$ can be written as a product

$$r = a_1 a_2 \dots a_m$$

Where: • q_i is irreducible for $i=1, \dots, m$

• (q_i) $i=1, \dots, m$

ideals are uniquely determined up to permutation.

Pf: 1A

$$q_1 q_2 \dots q_m = q'_1 q'_2 \dots q'_n$$

We want to show:

(a) $m=n$

(b) After reindexing, we have $(q_i) = (q'_i)$

$$(q_2) = (q'_2)$$

$\vdots$

$$(q_m) = (q'_m)$$

Do this by induction on m

Base-case: $m=1$

$$q_1 = q'_1 q'_2 \dots q'_m$$

$\Rightarrow$ Irreducibility of z_1
tells me $n=1$ and $z_1 = z_1'$.

Inductive step:

$$a_1 a_2 \dots a_m = z_1' z_2' \dots z_n'$$

$$\Rightarrow z_1' z_2' \dots z_n' \in (z_1)$$

$\Rightarrow$ After reindexing
 z_1 is prime $z_1' \in (z_1)$

$$\Rightarrow z_1' = z_1 b$$

$\Rightarrow$ z_1' is irreducible b is invertible

$$\Rightarrow (z_1') = (z_1)$$

we now have

$$\begin{aligned} a_1 a_2 \dots a_m &= (z_1 b) z_2' \dots z_n' \\ &= z_1 b z_2' \dots z_n' \end{aligned}$$

$\leftarrow$ grouping

$$\Rightarrow a_1(a_2 \cdots a_m - \underbrace{b a_2' \cdots a_n'}_{=0}) = 0$$

$$\Rightarrow \underbrace{a_2 \cdots a_m}_{m-1 \text{ factors}} = (\underbrace{b a_2' \cdots a_n'}_{n-1 \text{ factors}})$$

So we can apply inductive hypothesis to get the uniqueness statement.

Defⁿ An integral domain R with these properties is called a **Unique Factorization Domain** or **UFD**

Proposition: Every PID is
 $\hookrightarrow$ UFD

Remark: Converse is very far
from true. $\mathbb{Z}[x]$ is $\hookrightarrow$ UFD
but not $\hookrightarrow$ PID. In fact,

$\mathbb{Z}[x_1, x_2, \dots, x_n]$ is $\hookrightarrow$ UFD
but not $\hookrightarrow$ PID.

P.f.: We already know
that every irreducible element
is prime in $\hookrightarrow$ PID.

Choose $r \in R$ non-invertible
non-zero

If r is not irreducible

$$r = r_1 s_1$$

non-invertible

If r_1, s_1 are irreducible, we're done.

Otherwise, assume r_1 is reducible

$$r_1 = r_2 s_2$$

non-invertible

So either process terminates in irreducibles, or we have produced an infinite increasing chain

$$(r) \leq (r_1) \leq (r_2) \leq (r_3) \leq \dots \leq (r_n) \leq \dots$$

Claim: The union of this increasing chain of ideals is also an ideal I .

$$\Rightarrow I = (t)$$

(R is a PID!)

$\Rightarrow$ If $t \in (r_n)$ then

all of the ideals $(r_1), (r_2), \dots$,
are contained in (r_n)

$\Rightarrow r_n$ is actually irreducible!!

Example:

$$2\left(\frac{1+\sqrt{-3}}{2}\right) = \left\{ a + \frac{(1+\sqrt{-3})}{2} b : a, b \in \mathbb{Z} \right\}$$